Transfer Pump



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Juice and pulp transfer pump stainless steel

What it is used for

A stainless-steel transfer pump moves juice, pulp, and other compatible food liquids between tanks or processing stages. Correct setup and hygienic operation help protect product quality and prevent contamination.

Main parts

- Stainless-steel pump housing and impeller
- Food-grade inlet and discharge connections
- Motor and drive assembly
- Suction hose and discharge hose or pipes
- Priming port or seal arrangement if fitted
- Control switch and flow-control valves

Before you start

1. Confirm the pump is approved for the product, temperature, viscosity, and required flow rate.
2. Inspect the pump housing, hoses, seals, clamps, and electrical connections for damage.
3. Verify that product-contact surfaces are clean and sanitized according to the food safety procedure.
4. Check that the suction line is connected securely and the discharge line leads to the correct destination.
5. Confirm that valves are in the required positions and the inlet is not blocked.
6. Prime the pump when required by its design and verify the motor rotation if specified.

How to use

1. Ensure the receiving tank has sufficient capacity for the expected transfer volume.
2. Open the required inlet and discharge valves according to the operating instructions.
3. Switch on the pump and observe the initial flow through the discharge line.
4. Adjust the flow using approved controls without exceeding the pump's operating limits.
5. Monitor the product for leakage, excessive foaming, unusual noise, vibration, or overheating.
6. Prevent the pump from running dry and maintain the required inlet supply.
7. Stop the pump before changing hoses, opening connections, or switching the product route.
8. When transfer is complete, stop the pump and close the valves in the recommended sequence.
9. Drain remaining product safely and perform the approved cleaning and sanitizing procedure.
10. Disconnect and store hoses and accessories hygienically after cleaning and inspection.

⚠ Safety

- Use only food-compatible hoses, seals, and fittings for product contact.
- Never open the pump housing or loosen a pressurized connection while the pump is operating.
- Keep electrical connections dry and use suitable earthing and electrical protection.
- Isolate power before clearing blockages, cleaning the pump, or performing maintenance.
- Avoid touching hot motor surfaces and keep hands away from rotating components.
- Follow food hygiene procedures and wear suitable protection when handling hot or acidic products.

Care & storage

- Clean and sanitize all product-contact parts after use and between incompatible products.
- Inspect impellers, seals, clamps, and hoses for wear, cracks, or contamination.
- Follow the manufacturer's instructions for lubrication and servicing of non-product-contact components.
- Store the pump and hoses clean, dry, and protected from dust and pests.

Common mistakes to avoid

- Running the pump without adequate priming can prevent flow or damage components.
- Using non-food-grade hoses or damaged seals can contaminate the product.
- Operating against a closed discharge valve can cause pressure buildup and overheating.
- Skipping cleaning between batches can cause microbial growth and flavour carryover.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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